

LINEAR ALGEBRA

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MATRICES

An $m \times n$ **matrix** A is a rectangular array of mn real or complex numbers arranged in m horizontal **rows** and n vertical **columns**:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & \cdots & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & \cdots & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \cdots & \cdots & \vdots \\ \vdots & \vdots & \cdots & \cdots & a_{ij} & \vdots \\ \vdots & \vdots & \cdots & \cdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & \cdots & \cdots & a_{mn} \end{bmatrix} \quad \begin{array}{l} \leftarrow i \text{th row} \\ \uparrow j \text{th column} \end{array} \quad (1)$$

The i th row of A is

$$[a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}] \quad (1 \leq i \leq m);$$

the j th column of A is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix} \quad (1 \leq j \leq n).$$

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \text{L} & a_{1n} \\ a_{21} & a_{22} & a_{23} & \text{L} & a_{2n} \\ a_{31} & a_{32} & a_{33} & \text{L} & a_{3n} \\ \text{M} & \text{M} & \text{M} & & \text{M} \\ a_{m1} & a_{m2} & a_{m3} & \text{L} & a_{mn} \end{bmatrix}_{m \times n} \in M_{m \times n}$$

(i, j) -th entry: a_{ij}

row: m

column: n

size: $m \times n$

We shall say that A is m by n (written as $m \times n$). If $m = n$, we say that A is a **square matrix of order n** , and that the numbers $a_{11}, a_{22}, \dots, a_{nn}$ form the **main diagonal** of A . We refer to the number a_{ij} , which is in the i th row and j th column of A , as the **i, j th element** of A , or the **(i, j) entry** of A , and we often write (1) as

$$A = [a_{ij}].$$

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1+i & 4i \\ 2-3i & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix}, \quad E = [3], \quad F = [-1 \quad 0 \quad 2].$$

Then A is a 2×3 matrix with $a_{12} = 2$, $a_{13} = 3$, $a_{22} = 0$, and $a_{23} = 1$; B is a 2×2 matrix with $b_{11} = 1+i$, $b_{12} = 4i$, $b_{21} = 2-3i$, and $b_{22} = -3$; C is a 3×1 matrix with $c_{11} = 1$, $c_{21} = -1$, and $c_{31} = 2$; D is a 3×3 matrix; E is a 1×1 matrix; and F is a 1×3 matrix. In D , the elements $d_{11} = 1$, $d_{22} = 0$, and $d_{33} = 2$ form the main diagonal. ■

i -th row vector

$$\mathbf{r}_i = [\mathbf{a}_{i1} \quad \mathbf{a}_{i2} \quad \mathbf{L} \quad \mathbf{a}_{in}]$$

row matrix

j -th column vector

$$\mathbf{c}_j = \begin{bmatrix} \mathbf{c}_{1j} \\ \mathbf{c}_{2j} \\ \mathbf{M} \\ \mathbf{c}_{mj} \end{bmatrix}$$

column matrix

Square matrix: if $m = n$ than matrix is called square matrix.

Diagonal matrix:

$$A = \text{diag}(d_1, d_2, \mathbf{L}, d_n) = \begin{bmatrix} d_1 & 0 & \mathbf{K} & 0 \\ 0 & d_2 & \mathbf{L} & 0 \\ \mathbf{M} & \mathbf{M} & \mathbf{O} & \mathbf{M} \\ 0 & 0 & \mathbf{L} & d_n \end{bmatrix} \in M_{n \times n}$$

Trace:

$$\text{If } A = [a_{ij}]_{n \times n}$$

$$\text{Then } \text{Tr}(A) = a_{11} + a_{22} + \mathbf{L} + a_{nn}$$

Ex:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\Rightarrow r_1 = [1 \ 2 \ 3], \ r_2 = [4 \ 5 \ 6]$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = [c_1 \ c_2 \ c_3]$$

$$\Rightarrow c_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \ c_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \ c_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Partitioned matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

submatrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 \end{bmatrix}$$

(Tabular Display of Data) The following matrix gives the airline distances between the indicated cities (in statute miles):

	London	Madrid	New York	Tokyo
London	0	785	3469	5959
Madrid	785	0	3593	6706
New York	3469	3593	0	6757
Tokyo	5959	6706	6757	0



(Production) Suppose that a manufacturer has four plants, each of which makes three products. If we let a_{ij} denote the number of units of product i made by plant

j in one week, then the 3×4 matrix

	Plant 1	Plant 2	Plant 3	Plant 4
Product 1	560	360	380	0
Product 2	340	450	420	80
Product 3	280	270	210	380

gives the manufacturer's production for the week. For example, plant 2 makes 270 units of product 3 in one week. ■

Internet search engines use matrices to keep track of the locations of information, the type of information at a location, keywords that appear in the information, and even the way websites link to one another. A large measure of the effectiveness of the search engine Google[®] is the manner in which matrices are used to determine which sites are referenced by other sites. That is, instead of directly keeping track of the information content of an actual web page or of an individual search topic, Google's matrix structure focuses on finding web pages that match the search topic, and then presents a list of such pages in the order of their "importance."

Suppose that there are n accessible web pages during a certain month. A simple way to view a matrix that is part of Google's scheme is to imagine an $n \times n$ matrix A , called the "connectivity matrix," that initially contains all zeros. To build the connections, proceed as follows. When you detect that website j links to website i , set entry a_{ij} equal to one. Since n is quite large, in the billions, most entries of the connectivity matrix A are zero. (Such a matrix is called sparse.)

Equal matrix:

$$\text{Let } A = [a_{ij}]_{m \times n}, \quad B = [b_{ij}]_{m \times n}$$

$$A = B \quad \text{if and only if} \quad a_{ij} = b_{ij} \quad \forall \quad 1 \leq i \leq m, \quad 1 \leq j \leq n$$

The matrices

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -3 & 4 \\ 0 & -4 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & w \\ 2 & x & 4 \\ y & -4 & z \end{bmatrix}$$

are equal if and only if $w = -1$, $x = -3$, $y = 0$, and $z = 5$.

Matrix Addition

Definition The *sum* of two same-sized matrices is their entry-by-entry sum. The *scalar multiple* of a matrix is the result of entry-by-entry scalar multiplication.

Matrix addition:

If $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$

Then $A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n}$

Let

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -1 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & -4 \end{bmatrix}.$$

Then

$$A + B = \begin{bmatrix} 1+0 & -2+2 & 3+1 \\ 2+1 & -1+3 & 4+(-4) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 4 \\ 3 & 2 & 0 \end{bmatrix}.$$

(Production) A manufacturer of a certain product makes three models, A, B, and C. Each model is partially made in factory F_1 in Taiwan and then finished in factory F_2 in the United States. The total cost of each product consists of the manufacturing cost and the shipping cost. Then the costs at each factory (in dollars) can be described by the 3×2 matrices F_1 and F_2 :

$$F_1 = \begin{array}{cc} \begin{array}{c} \text{Manufacturing} \\ \text{cost} \end{array} & \begin{array}{c} \text{Shipping} \\ \text{cost} \end{array} \\ \left[\begin{array}{cc} 32 & 40 \\ 50 & 80 \\ 70 & 20 \end{array} \right] & \begin{array}{l} \text{Model A} \\ \text{Model B} \\ \text{Model C} \end{array} \end{array}$$

$$F_2 = \begin{array}{cc} \begin{array}{c} \text{Manufacturing} \\ \text{cost} \end{array} & \begin{array}{c} \text{Shipping} \\ \text{cost} \end{array} \\ \left[\begin{array}{cc} 40 & 60 \\ 50 & 50 \\ 130 & 20 \end{array} \right] & \begin{array}{l} \text{Model A} \\ \text{Model B} \\ \text{Model C} \end{array} \end{array}.$$

The matrix $F_1 + F_2$ gives the total manufacturing and shipping costs for each product. Thus the total manufacturing and shipping costs of a model C product are \$200 and \$40, respectively. ■

!!!!We now make the convention that when $A+B$ is written, both A and B are of the same size.

Scalar Multiplication

Scalar multiplication:

If $A = [a_{ij}]_{m \times n}$, c : scalar

Then $cA = [ca_{ij}]_{m \times n}$

We have

$$\begin{aligned} -2 \begin{bmatrix} 4 & -2 & -3 \\ 7 & -3 & 2 \end{bmatrix} &= \begin{bmatrix} (-2)(4) & (-2)(-2) & (-2)(-3) \\ (-2)(7) & (-2)(-3) & (-2)(2) \end{bmatrix} \\ &= \begin{bmatrix} -8 & 4 & 6 \\ -14 & 6 & -4 \end{bmatrix}. \end{aligned}$$



Matrix subtraction:

$$A - B = A + (-1)B$$

Let

$$A = \begin{bmatrix} 2 & 3 & -5 \\ 4 & 2 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 5 & -2 \end{bmatrix}.$$

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Then

$$A - B = \begin{bmatrix} 2-2 & 3+1 & -5-3 \\ 4-3 & 2-5 & 1+2 \end{bmatrix} = \begin{bmatrix} 0 & 4 & -8 \\ 1 & -3 & 3 \end{bmatrix}.$$

Find the linear combinations of following matrix

$$3 \begin{bmatrix} 0 & -3 & 5 \\ 2 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 5 & 2 & 3 \\ 6 & 2 & 3 \\ -1 & -2 & 3 \end{bmatrix},$$

Find the linear combinations of following matrix

$$3 \begin{bmatrix} 0 & -3 & 5 \\ 2 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 5 & 2 & 3 \\ 6 & 2 & 3 \\ -1 & -2 & 3 \end{bmatrix},$$

Answer:

$$\begin{bmatrix} -\frac{5}{2} & -10 & \frac{27}{2} \\ 3 & 8 & \frac{21}{2} \\ \frac{7}{2} & -5 & -\frac{21}{2} \end{bmatrix}$$

Find the linear combinations of following matrix

$$2 \begin{bmatrix} 3 & -2 \end{bmatrix} - 3 \begin{bmatrix} 5 & 0 \end{bmatrix} + 4 \begin{bmatrix} -2 & 5 \end{bmatrix},$$

Find the linear combinations of following matrix

$$2 \begin{bmatrix} 3 & -2 \end{bmatrix} - 3 \begin{bmatrix} 5 & 0 \end{bmatrix} + 4 \begin{bmatrix} -2 & 5 \end{bmatrix},$$

Answer:

$$\begin{bmatrix} -17 & 16 \end{bmatrix}$$

Find the linear combinations of following matrix

$$-0.5 \begin{bmatrix} 1 \\ -4 \\ -6 \end{bmatrix} + 0.4 \begin{bmatrix} 0.1 \\ -4 \\ 0.2 \end{bmatrix}.$$

Find the linear combinations of following matrix

$$-0.5 \begin{bmatrix} 1 \\ -4 \\ -6 \end{bmatrix} + 0.4 \begin{bmatrix} 0.1 \\ -4 \\ 0.2 \end{bmatrix}.$$

Answer:

$$\begin{bmatrix} -0.46 \\ 0.4 \\ 3.08 \end{bmatrix}$$

Let

$$\mathbf{p} = \begin{bmatrix} 18.95 \\ 14.75 \\ 8.60 \end{bmatrix}$$

be a 3-vector that represents the current prices of three items at a store. Suppose that the store announces a sale so that the price of each item is reduced by 20%.

- (a) Determine a 3-vector that gives the price changes for the three items.
- (b) Determine a 3-vector that gives the new prices of the items.

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- (a) Determine a 3-vector that gives the price changes for the three items.
- (b) Determine a 3-vector that gives the new prices of the items.

Solution

- (a) Since each item is reduced by 20%, the 3-vector

$$-0.20\mathbf{p} = \begin{bmatrix} (-0.20)18.95 \\ (-0.20)14.75 \\ (-0.20)8.60 \end{bmatrix} = \begin{bmatrix} -3.79 \\ -2.95 \\ -1.72 \end{bmatrix} = - \begin{bmatrix} 3.79 \\ 2.95 \\ 1.72 \end{bmatrix}$$

gives the price changes for the three items.

- (b) The new prices of the items are given by the expression

$$\mathbf{p} - 0.20\mathbf{p} = \begin{bmatrix} 18.95 \\ 14.75 \\ 8.60 \end{bmatrix} - \begin{bmatrix} 3.79 \\ 2.95 \\ 1.72 \end{bmatrix} = \begin{bmatrix} 15.16 \\ 11.80 \\ 6.88 \end{bmatrix}.$$

Observe that this expression can also be written as

$$\mathbf{p} - 0.20\mathbf{p} = 0.80\mathbf{p}.$$

Matrix Multiplication

In this section we introduce the operation of matrix multiplication. Unlike matrix addition, matrix multiplication has some properties that distinguish it from multiplication of real numbers.

The **dot product**, or **inner product**, of the n -vectors in R^n

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

is defined as

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^n a_i b_i.$$

The dot product of

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 3 \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

The dot product of

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 3 \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

is

$$\mathbf{u} \cdot \mathbf{v} = (1)(2) + (-2)(3) + (3)(-2) + (4)(1) = -6.$$

Question:

Let $\mathbf{a} = \begin{bmatrix} x \\ 2 \\ 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$. If $\mathbf{a} \cdot \mathbf{b} = -4$, find x .

Question:

Let $\mathbf{a} = \begin{bmatrix} x \\ 2 \\ 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$. If $\mathbf{a} \cdot \mathbf{b} = -4$, find x .

Solution

We have

$$\mathbf{a} \cdot \mathbf{b} = 4x + 2 + 6 = -4$$

$$4x + 8 = -4$$

$$x = -3.$$

- Determining if two vectors are Orthogonal – Two conformable vectors **a** and **b** are *orthogonal* iff

$$\mathbf{a}^t \mathbf{b} = 0$$

Example: Suppose we have

then $\mathbf{a} = \begin{bmatrix} -7 & 10 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 & 0.5 & -2 \end{bmatrix}$

$$\mathbf{a} \mathbf{b}^t = \begin{bmatrix} -7 & 10 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \\ -2 \end{bmatrix} = -7(1) + 10(0.5) - 1(-2) = 0$$

DEFINITION (Product of Matrices):

If $A = [a_{ij}]$ $m \times p$ matrix and $B = [b_{ij}]$ is

$p \times n$ matrix then the product of A and B , denoted by AB is the $m \times n$ matrix $C = [c_{ij}]$ defined by

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}$$

$$(1 \leq i \leq m, 1 \leq j \leq n)$$

Matrix multiplication:

$$\begin{bmatrix} a_{11} & a_{12} & \text{L} & a_{1n} \\ \text{M} & \text{M} & & \text{M} \\ a_{i1} & a_{i2} & \text{L} & a_{in} \\ \text{M} & \text{M} & & \text{M} \\ a_{n1} & a_{n2} & \text{L} & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & \text{L} & b_{1j} & \text{L} & b_{1n} \\ b_{21} & \text{M} & b_{2j} & \text{L} & b_{2n} \\ \text{M} & \text{M} & \text{M} & & \text{M} \\ b_{n1} & \text{L} & b_{nj} & \text{L} & b_{nn} \end{bmatrix} = \begin{bmatrix} c_{i1} & c_{i2} & \text{L} & c_{ij} & \text{L} & c_{in} \end{bmatrix}$$

Notes: (1) $A+B = B+A$,

(2) $AB \neq BA$

Matrix Multiplication

An Example

- If we have $\mathbf{A} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

then $\mathbf{AB} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix} = \begin{bmatrix} 30 & 66 \\ 36 & 81 \\ 42 & 96 \end{bmatrix}$

where

$$\begin{aligned} c_{11} &= a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = 1(1) + 4(2) + 7(3) = 30 \\ c_{12} &= a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} = 1(4) + 4(5) + 7(6) = 66 \\ c_{21} &= a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} = 2(1) + 5(2) + 8(3) = 36 \\ c_{22} &= a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} = 2(4) + 5(5) + 8(6) = 81 \\ c_{31} &= a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} = 3(1) + 6(2) + 9(3) = 42 \\ c_{32} &= a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} = 3(4) + 6(5) + 9(6) = 96 \end{aligned}$$

Be careful!!!!

If A is a 2×3 matrix and B is a 3×4 matrix, then AB is a 2×4 matrix while BA is undefined. ■

Let A be 2×3 and let B be 3×2 . Then AB is 2×2 while BA is 3×3 . ■

Let

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 2 & 3 \\ -2 & 2 \end{bmatrix} \quad \text{while} \quad BA = \begin{bmatrix} 1 & 7 \\ -1 & 3 \end{bmatrix}.$$

Thus $AB \neq BA$. ■

Let

$$A = \begin{bmatrix} 2 & -1 & -3 \\ 4 & 2 & -2 \end{bmatrix} \quad \text{and} \quad \mathbf{c} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Then the product $A\mathbf{c}$, written as a linear combination of the columns of A , is

$$A\mathbf{c} = \begin{bmatrix} 2 & -1 & -3 \\ 4 & 2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} - 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 4 \begin{bmatrix} -3 \\ -2 \end{bmatrix} = \begin{bmatrix} -5 \\ -6 \end{bmatrix}. \quad \blacksquare$$

Let

$$A = \begin{bmatrix} 1 & x & 3 \\ 2 & -1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 \\ 4 \\ y \end{bmatrix}.$$

If $AB = \begin{bmatrix} 12 \\ 6 \end{bmatrix}$, find x and y .

Matrix form of a system of linear equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \text{L} + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \text{L} + a_{2n}x_n = b_2 \\ \text{M} \\ a_{m1}x_1 + a_{m2}x_2 + \text{L} + a_{mn}x_n = b_m \end{cases}$$



$$\begin{bmatrix} a_{11} & a_{12} & \text{L} & a_{1n} \\ a_{21} & a_{22} & \text{L} & a_{2n} \\ \text{M} & \text{M} & \text{M} & \text{M} \\ a_{m1} & a_{m2} & \text{L} & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \text{M} \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \text{M} \\ b_m \end{bmatrix}$$

$$\begin{matrix} \parallel & \parallel & \parallel \\ \mathbf{A} & \mathbf{x} & \mathbf{b} \end{matrix}$$

m linear equations



Single matrix equation

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

$$\begin{matrix} m \times n & n \times 1 & m \times 1 \end{matrix}$$

a linear combination of the column vectors of matrix A:

$$A = \begin{bmatrix} a_{11} & a_{12} & \mathbf{L} & a_{1n} \\ a_{21} & a_{22} & \mathbf{L} & a_{2n} \\ \mathbf{M} & \mathbf{M} & \mathbf{M} & \mathbf{M} \\ a_{m1} & a_{m2} & \mathbf{L} & a_{mn} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & \mathbf{L} & c_n \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \mathbf{M} \\ x_n \end{bmatrix}$$

$$\Rightarrow Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \mathbf{L} + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \mathbf{L} + a_{2n}x_n \\ \mathbf{M} \\ a_{m1}x_1 + a_{m2}x_2 + \mathbf{L} + a_{mn}x_n \end{bmatrix}_{m \times 1} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \mathbf{M} \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{22} \\ \mathbf{M} \\ a_{m2} \end{bmatrix} + \mathbf{L} + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \mathbf{M} \\ a_{mn} \end{bmatrix}$$

$\parallel$
 c_1

$\parallel$
 c_2

$\parallel$
 c_n

linear combination of column
vectors of A

Properties of Matrix Operations

Three basic matrix operators:

- (1) matrix addition
- (2) scalar multiplication
- (3) matrix multiplication

Zero matrix: $0_{m \times n}$

Identity matrix of order n : I_n

Properties of matrix addition and scalar multiplication:

If $A, B, C \in M_{m \times n}$, c, d : scalar

Then (1) $A+B = B + A$

Commutative law for addition

(2) $A + (B + C) = (A + B) + C$

Associative law for addition

(3) $(cd) A = c (dA)$

Associative law for scalar multiplication

(4) $1A = A$

Unit element for scalar multiplication

(5) $c(A+B) = cA + cB$

Distributive law 1 for scalar multiplication

(6) $(c+d) A = cA + dA$

Distributive law 2 for scalar multiplication

Properties of zero matrices:

If $A \in M_{m \times n}$, $c : \text{scalar}$

Then (1) $A + 0_{m \times n} = A$ the additive identity for all $m \times n$ matrices

(2) $A + (-A) = 0_{m \times n}$ the additive inverse of A

(3) $cA = 0_{m \times n} \Rightarrow c = 0 \text{ or } A = 0_{m \times n}$

Properties of matrix multiplication:

Theorem If F , G , and H are matrices, and the matrix products are defined, then the product is associative $(FG)H = F(GH)$ and distributes over matrix addition $F(G + H) = FG + FH$ and $(G + H)F = GF + HF$.

Properties of the multiplicative identity matrix:

If $A \in M_{m \times n}$

Then (1) $AI_n = A$

(2) $I_m A = A$

Transpose of a matrix:

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} & \text{L} & a_{1n} \\ a_{21} & a_{22} & \text{L} & a_{2n} \\ \text{M} & \text{M} & \text{M} & \text{M} \\ a_{m1} & a_{m2} & \text{L} & a_{mn} \end{bmatrix} \in M_{m \times n}$$

$$\text{Then } A^T = \begin{bmatrix} a_{11} & a_{21} & \text{L} & a_{m1} \\ a_{12} & a_{22} & \text{L} & a_{m2} \\ \text{M} & \text{M} & \text{M} & \text{M} \\ a_{1n} & a_{2n} & \text{L} & a_{mn} \end{bmatrix} \in M_{n \times m}$$

Properties of transposes:

$$(1) (A^T)^T = A$$

$$(2) (A + B)^T = A^T + B^T$$

$$(3) (cA)^T = c(A^T)$$

$$(4) (AB)^T = B^T A^T$$

- Also note that
 - The transpose of a product is equal to the product of the transposes in reverse order $\rightarrow (ABC)^t = C^t B^t A^t$

Find the transpose of following matrices

$$A = \begin{bmatrix} 4 & -2 & 3 \\ 0 & 5 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 2 & -4 \\ 3 & -1 & 2 \\ 0 & 4 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 5 & 4 \\ -3 & 2 \\ 2 & -3 \end{bmatrix},$$
$$D = [3 \quad -5 \quad 1], \quad E = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

Answer:

$$A^T = \begin{bmatrix} 4 & 0 \\ -2 & 5 \\ 3 & -2 \end{bmatrix}, \quad B^T = \begin{bmatrix} 6 & 3 & 0 \\ 2 & -1 & 4 \\ -4 & 2 & 3 \end{bmatrix},$$

$$C^T = \begin{bmatrix} 5 & -3 & 2 \\ 4 & 2 & -3 \end{bmatrix}, \quad D^T = \begin{bmatrix} 3 \\ -5 \\ 1 \end{bmatrix}, \quad \text{and} \quad E^T = [2 \quad -1 \quad 3].$$



Symmetric matrix:

A square matrix A is symmetric if $A = A^T$

Skew-symmetric matrix:

A square matrix A is skew-symmetric if $A^T = -A$

Note: AA^T is symmetric

Pf: $(AA^T)^T = (A^T)^T A^T = AA^T$
 $\therefore AA^T$ is symmetric

Real number:

$$ab = ba \quad (\text{Commutative law for multiplication})$$

Matrix:

$$\underset{m \times n}{\mathbf{A}} \underset{n \times p}{\mathbf{B}} \neq \mathbf{B} \mathbf{A}$$

Three situations of the non-commutativity may occur :

- (1) If $m \neq p$, then AB is defined, BA is undefined.
- (2) If $m = p, m \neq n$, then $AB \in M_{m \times n}$ $BA \in M_{n \times n}$ (Sizes are not the same)
- (3) If $m = p = n$, then $AB \in M_{m \times n}$ $BA \in M_{m \times m}$
(Sizes are the same, but matrices are not equal)

Example :

Show that AB and BA are not equal for the matrices.

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$$

Sol:

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 4 & -4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 4 & -2 \end{bmatrix}$$

Real number:

$$ac = bc, c \neq 0$$

$$\Rightarrow a = b \quad (\text{Cancellation law})$$

Matrix: Be careful!!!!

$$AC = BC \quad C \neq 0$$

(1) **If C is invertible** (i.e., C^{-1} exists), then $A = B$

(2) If C is not invertible, then $A \neq B$ (Cancellation is not valid)

Example: **An example in which cancellation is not valid**
Show that $AC=BC$

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

**C is noninvertible,
(i.e., row 1 and row 2
are not independent)**

Sol:

$$AC = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

So $AC = BC$

But $A \neq B$

Exercises:

Let

1)

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 6 & -5 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ -3 \\ 5 \end{bmatrix},$$

and

$$C = \begin{bmatrix} 7 & 3 & 2 \\ -4 & 3 & 5 \\ 6 & 1 & -1 \end{bmatrix}.$$

(a) What is a_{12} , a_{22} , a_{23} ?

(b) What is b_{11} , b_{31} ?

(c) What is c_{13} , c_{31} , c_{33} ?

2)

If

$$\begin{bmatrix} a+b & c+d \\ c-d & a-b \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 10 & 2 \end{bmatrix},$$

find a , b , c , and d .

3)

If

$$\begin{bmatrix} a+2b & 2a-b \\ 2c+d & c-2d \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 4 & -3 \end{bmatrix},$$

find a , b , c , and d .

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix},$$

$$4) \quad C = \begin{bmatrix} 3 & -1 & 3 \\ 4 & 1 & 5 \\ 2 & 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & -2 \\ 2 & 4 \end{bmatrix},$$

$$E = \begin{bmatrix} 2 & -4 & 5 \\ 0 & 1 & 4 \\ 3 & 2 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} -4 & 5 \\ 2 & 3 \end{bmatrix}.$$

If possible, compute the indicated linear combination:

(a) $C + E$ and $E + C$ (b) $A + B$

(c) $D - F$ (d) $-3C + 5O$

(e) $2C - 3E$ (f) $2B + F$

If possible, compute the indicated linear combination:

(a) $3D + 2F$ (b) $3(2A)$ and $6A$

(c) $3A + 2A$ and $5A$

(d) $2(D + F)$ and $2D + 2F$

(e) $(2 + 3)D$ and $2D + 3D$

(f) $3(B + D)$

If possible, compute the following:

(a) A^T and $(A^T)^T$

(b) $(C + E)^T$ and $C^T + E^T$

(c) $(2D + 3F)^T$ (d) $D - D^T$

(e) $2A^T + B$ (f) $(3D - 2F)^T$

- 5) Is the matrix $\begin{bmatrix} 4 & 1 \\ 0 & -3 \end{bmatrix}$ a linear combination of the matrices $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$? Justify your answer.

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 6 & -2 & 3 \\ 5 & 2 & 4 \end{bmatrix} \quad \text{and} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- 6) If λ is a real number, compute $\lambda I_3 - A$.
If A is an $n \times n$ matrix, what are the entries on the main diagonal of $A - A^T$? Justify your answer.
- 7) Let the $n \times n$ matrix A be equal to A^T . Briefly describe the pattern of the entries in A .
If $\mathbf{x}$ is an n -vector, show that $\mathbf{x} + \mathbf{0} = \mathbf{x}$.

Commutative Matrices

- Any two square matrices **A** and **B** such that **AB** = **BA** are said to *commute*.

Note that it is easy to show that any square matrix **A** commutes with both itself and with a conformable identity matrix **I**.

Anti-Commutative Matrices

- Any two square matrices **A** and **B** such that **AB** = -**BA** are said to *anti-commute*.

Periodic Matrices

- Any matrix $\mathbf{A}$ such that $\mathbf{A}^{k+1} = \mathbf{A}$ is said to be of *period* k .

Idempotent Matrices

- Any matrix **A** such that $\mathbf{A}^2 = \mathbf{A}$ is said to be of *idempotent*.

Nilpotent Matrices

- Any matrix **A** such that $\mathbf{A}^p = \mathbf{0}$ where p is a positive integer is said to be of *nilpotent*.

Note that if p is the least positive integer such that $\mathbf{A}^p = \mathbf{0}$, then A is said to be nilpotent of index p .

Orthogonal Matrices

- Any square matrix **A** with rows (considered as vectors) are mutually perpendicular and have unit lengths, i.e.,
$$A^t A = I$$

$$A^t = A^{-1}$$

Note that **A** is orthogonal iff

